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ASSIGNMENT - 1

Installation of Hadoop on a Single Node Cluster

Here is the step-by-step procedure of Hadoop installation on WSL (Ubuntu).

Step 1: Install Java 8 version: Hadoop requires Java to run and manage its nodes. We installed OpenJDK 8.

```
kanishka@LAPTOP-SBH634N3:~$ java -version
openjdk version "1.8.0_482"
OpenJDK Runtime Environment (build 1.8.0_482-8u482-ga~us1-0ubuntu1~24.04-b08)
OpenJDK 64-Bit Server VM (build 25.482-b08, mixed mode)
kanishka@LAPTOP-SBH634N3:~$
```

Step 2: Download and Extract Hadoop (version 3.3.6): Downloaded Hadoop using wget and extracted the files using tar -xzvf hadoop-3.3.6.tar.gz .

```
kanishka@LAPTOP-SBH634N3:~$ wget https://downloads.apache.org/hadoop/common/hadoop-3.3.6/hadoop-3.3.6.tar.gz
--2026-04-22 07:07:11--  https://downloads.apache.org/hadoop/common/hadoop-3.3.6/hadoop-3.3.6.tar.gz
Resolving downloads.apache.org (downloads.apache.org)... 135.181.214.104, 88.99.208.237, 2a01:4f8:10a:39da::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|135.181.214.104|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 730107476 (696M) [application/x-gzip]
Saving to: ‘hadoop-3.3.6.tar.gz’

hadoop-3.3.6.tar.gz  100%[===================>] 696.28M  1016KB/s    in 6m 35s

2026-04-22 07:13:52 (1.76 MB/s) - ‘hadoop-3.3.6.tar.gz’ saved [730107476/730107476]

kanishka@LAPTOP-SBH634N3:~$
```

Step 3: Configure Environment Variables: Moved Hadoop to /usr/local/hadoop and added the environment variables for JAVA_HOME and HADOOP_HOME to the ~/.bashrc file.

```
  GNU nano 7.2                                    core-site.xml *
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
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  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

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  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
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-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

Step 4: Configure Hadoop (Single Node) XML Files: Edited the four core configuration files (core-site.xml, hdfs-site xml, mapred-site xml, yarn-site xml) and hardcoded the Java path in hadoop-env.sh

```
kanishka@LAPTOP-SBH634N3:~$ cd /usr/local/hadoop/etc/hadoop
kanishka@LAPTOP-SBH634N3:/usr/local/hadoop/etc/hadoop$ nano core-site.xml
kanishka@LAPTOP-SBH634N3:/usr/local/hadoop/etc/hadoop$ nano hdfs-site.xml
```

```
kanishka@LAPTOP-SBH634N3:/usr/local/hadoop/etc/hadoop$ nano mapred-site.xml
kanishka@LAPTOP-SBH634N3:/usr/local/hadoop/etc/hadoop$ nano yarn-site.xml
kanishka@LAPTOP-SBH634N3:/usr/local/hadoop/etc/hadoop$ nano hadoop-env.sh
kanishka@LAPTOP-SBH634N3:/usr/local/hadoop/etc/hadoop$ echo $JAVA_HOME
```

Step 5: Configure SSH for Localhost: Generated RSA keys and authorized them to allow Hadoop to securely communicate with its components without "Connection refused" errors.

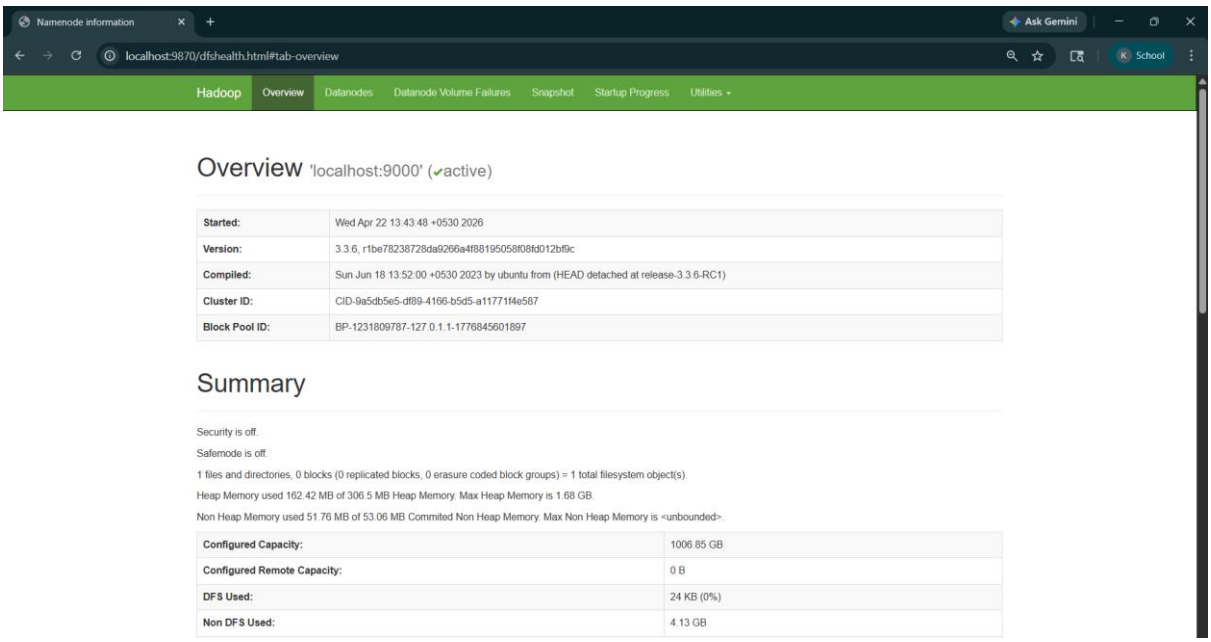
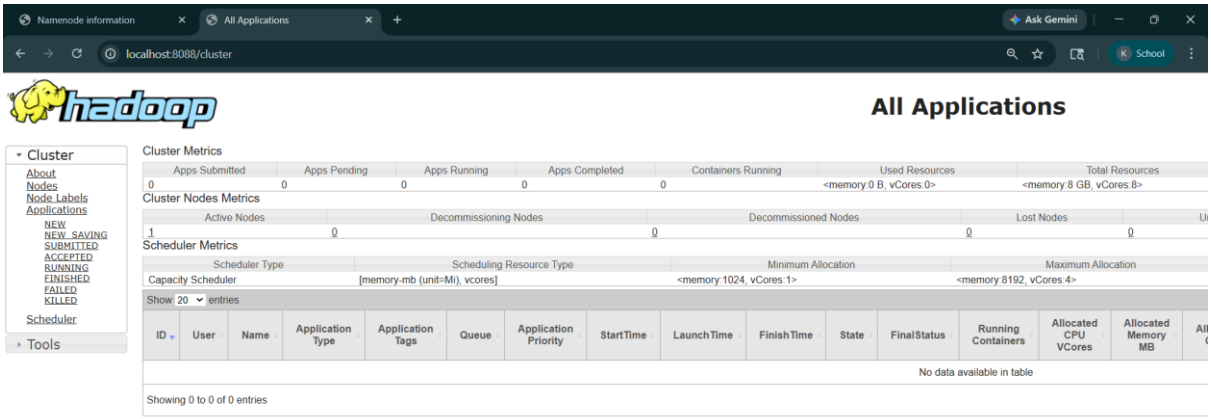
```
kanishka@LAPTOP-SBH634N3:/$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
kanishka@LAPTOP-SBH634N3:/$ ssh localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:r0Mt5fdHrZA+hkZr/fFfC52cHPev+0dX2t2aEwkRRZs.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
```

Step 6: Format NameNode: Formatted the file system for the first-time setup using hdfs namenode -format .

```
kanishka@LAPTOP-SBH634N3:~$ hdfs namenode -format
WARNING: /usr/local/hadoop/logs does not exist. Creating.
2026-04-22 08:13:20,304 INFO namenode.NameNode: STARTUP_MSG:
/************************************************************
STARTUP_MSG: Starting NameNode
STARTUP_MSG:   host = LAPTOP-SBH634N3/127.0.1.1
STARTUP_MSG:   args = [-format]
STARTUP_MSG:   version = 3.3.6
STARTUP_MSG:   classpath = /usr/local/hadoop/etc/hadoop:/usr/local/hadoop/share/hadoop/common/lib/ker
b-core-1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/protobuf-java-2.5.0.jar:/usr/local/hadoop/
share/hadoop/common/lib/netty-codec-stomp-4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/
netty-handler-4.1.89.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/kerb-common-1.0.1.jar:/usr/l
ocal/hadoop/share/hadoop/common/lib/jaxb-api-2.2.11.jar:/usr/local/hadoop/share/hadoop/common/lib/com
```

Step 7: Start Hadoop Services: Started the distributed file system and resource manager using start-dfs.sh and start-yarn.sh , then verified the running processes using jps .

```
kanishka@LAPTOP-SBH634N3:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [LAPTOP-SBH634N3]
LAPTOP-SBH634N3: Warning: Permanently added 'laptop-sbh634n3' (ED25519) to the list of known hosts.
kanishka@LAPTOP-SBH634N3:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
kanishka@LAPTOP-SBH634N3:~$ jps
3507 ResourceManager
3285 SecondaryNameNode
3063 DataNode
3623 NodeManager
3993 Jps
2942 NameNode
kanishka@LAPTOP-SBH634N3:~$
```

Step 8: Check the Web Interface : Opened the browser in Windows to verify the HDFS UI localhost : 9870) and Yarn UI (localhost:8088).

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